

# Machine Learning and Mobile Applications for Emotion Recognition in Children with Autism Spectrum Disorder: A Systematic Literature Review

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## Citation

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## REVIEW TITLE AND BASIC DETAILS

### Review title

Machine Learning and Mobile Applications for Emotion Recognition in Children with Autism Spectrum Disorder: A Systematic Literature Review

### Condition or domain being studied

*Autism Spectrum Disorder; Emotional State ; Problematic Behavior In Children*

This systematic review examines machine learning technologies and mobile applications designed for emotion recognition in children with Autism Spectrum Disorder (ASD). The focus is on pediatric populations (ages 0-18 years) and technologies applicable to resource-limited settings, particularly in Latin American contexts. The review synthesizes evidence on neural network architectures, performance metrics, datasets, and implementation barriers for emotion recognition systems deployed on mobile devices.

### Rationale for the review

Emotion recognition deficits represent a critical barrier to social development in children with Autism Spectrum Disorder (ASD), yet evidence-based technological interventions remain scarce, particularly in resource-limited settings. While machine learning technologies, especially convolutional neural networks, have demonstrated promising capabilities for facial emotion recognition, existing literature presents substantial methodological heterogeneity regarding architectures, performance metrics, datasets, and target populations. No systematic review has specifically synthesized evidence on mobile-based machine learning applications for emotion recognition in children with ASD, considering technical characteristics, implementation barriers, and accessibility requirements for resource-constrained environments. This review addresses this gap by systematically analyzing neural network architectures, performance benchmarks, available datasets, and methodological limitations identified in current literature. Findings will establish the current state-of-the-art and identify opportunities for developing culturally adapted, accessible therapeutic tools validated in Latin American contexts, informing future research directions in this underserved area.

## Review objectives

This systematic review aims to synthesize evidence on machine learning technologies applied to emotion recognition in children with Autism Spectrum Disorder (ASD), specifically addressing six research questions: (1) Which machine learning architectures and models demonstrate greatest effectiveness for emotion recognition and ASD detection in pediatric populations? (2) What performance metrics are consistently employed to evaluate these algorithms in clinical contexts? (3) What accuracy, precision, recall, F1-score, and AUC values do models achieve across different methodological configurations? (4) Which datasets are utilized for model training and validation, and what are their representativeness limitations? (5) What input variables and facial features do emotion recognition systems employ? (6) What technical and methodological limitations impede implementation of mobile-based systems in resource-limited contexts? The review will establish current state-of-the-art and identify gaps for developing accessible, culturally adapted therapeutic tools.

## Keywords

Emotion recognition; Facial expression recognition; Convolutional neural networks; Deep learning; Mobile applications; Smartphone; Assistive technology; Pediatric population; Neural networks; Digital interventions; Serious games; Resource-limited settings

## Country

Peru

## ELIGIBILITY CRITERIA

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### Population

#### *Included*

Children and adolescents aged 0-18 years diagnosed with Autism Spectrum Disorder (ASD) according to recognized diagnostic criteria (DSM-5, ICD-10, or equivalent). Studies must focus on emotion recognition or ASD detection using machine learning approaches with mobile applications or smartphone-based technologies.

#### *Excluded*

Adults over 18 years of age. Studies focusing exclusively on non-facial modalities (e.g., voice-only, physiological signals without facial analysis). Studies without empirical validation or quantitative performance metrics.

### Intervention(s) or exposure(s)

#### *Included*

*Smartphone app; Machine learning; Assistive technology*

Machine learning technologies and mobile applications designed for facial emotion recognition in children with ASD. This includes: convolutional neural network (CNN) architectures, deep learning models, smartphone-based applications, serious games, and digital therapeutic interventions. Systems must employ emotion recognition algorithms and be deployable on mobile devices or adaptable to mobile platforms.

#### *Excluded*

Desktop-only applications not adaptable to mobile platforms. Interventions focusing exclusively on non-facial modalities without facial expression analysis. Review articles, systematic reviews, or meta-analyses. Studies without quantitative performance metrics or empirical validation.

### Comparator(s) or control(s)

#### *Included*

Alternative machine learning architectures (e.g., traditional ML vs. deep learning, different CNN architectures such as ResNet vs. VGGFace vs. MobileNet), ensemble versus non-ensemble models, unimodal versus multimodal approaches, and different preprocessing or optimization techniques (SMOTE, data augmentation, k-fold cross-validation).

***Excluded***

Non-technological interventions, pharmacological treatments, behavioral therapies without technological components.

**Study design**

Only nonrandomized study types will be included.

***Included***

Technology development studies, validation studies, diagnostic accuracy studies, cross-sectional studies, observational studies, and machine learning model development and validation studies published in peer-reviewed journals or conference proceedings. Studies must report quantitative performance metrics for machine learning algorithms applied to emotion recognition or ASD detection. Both comparative studies (comparing different ML architectures) and non-comparative studies (reporting single model performance) will be included.

***Excluded***

Systematic reviews, meta-analyses, narrative reviews, editorials, commentaries, case reports, qualitative studies without quantitative validation, and studies without empirical data or performance metrics.

**Context**

Studies must employ machine learning technologies applicable to mobile devices, smartphone-based applications, or technologies adaptable to mobile platforms for emotion recognition or ASD detection. Studies may be conducted in any geographical setting but must report on technologies designed for or tested in real-world clinical, educational, or home environments. Particular emphasis on technologies suitable for resource-limited settings, including those with limited internet connectivity or requiring offline functionality. Studies conducted exclusively in controlled laboratory settings without consideration for real-world implementation will be included but noted as having limited ecological validity. No restrictions based on country of origin, but datasets and validation must involve pediatric populations (ages 0-18 years).

## **TIMELINE OF THE REVIEW**

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**Date of first submission to PROSPERO**

10 December 2025

**Review timeline**

Start date: 15 October 2025. End date: 31 March 2026.

**Date of registration in PROSPERO**

11 December 2025

## **AVAILABILITY OF FULL PROTOCOL**

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**Availability of full protocol**

A full protocol has been written but is not available because:

*The full protocol is incorporated within a manuscript currently under review by thesis advisors prior to journal submission. The protocol will be made publicly available upon publication of the systematic review*

## SEARCHING AND SCREENING

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### Search for unpublished studies

Only published studies will be sought.

### Main bibliographic databases that will be searched

The main database to be searched is *Scopus*.

### Other important or specialist databases that will be searched

IEEE Xplore Digital Library, Web of Science Core Collection

### Search language restrictions

The review will only include studies published in English.

### Search date restrictions

Databases will be searched for articles published from 1 January 2018 and before by 1 October 2025.

### Other methods of identifying studies

No other methods will be used.

### Link to search strategy

A full search strategy is available in the full protocol as described in the *Availability of full protocol* section

### Selection process

Studies will be screened independently by at least two people (or person/machine combination) with a process to resolve differences.

### Other relevant information about searching and screening

Inter-rater reliability was monitored using Cohen's kappa coefficient ( $\kappa = 0.82$  for title/abstract screening;  $\kappa = 0.89$  for full-text review, indicating substantial to excellent agreement). All searches were conducted on October 15, 2025. References were managed using Mendeley Desktop v1.19.8 with automated deduplication complemented by manual verification. Complete documentation of all PRISMA phases with exclusion justifications is maintained in supplementary materials.

## DATA COLLECTION PROCESS

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### Data extraction from published articles and reports

Data will be extracted independently by at least two people (or person/machine combination) with a process to resolve differences.

Authors will not be contacted for further information.

### Study risk of bias or quality assessment

Risk of bias will be assessed using:

Custom quality assessment framework adapted from best practices for ML model reporting and validation, evaluating: (1) architecture description clarity and reproducibility, (2) code/model availability, (3) validation strategy robustness, (4) performance metrics comprehensiveness, (5) error analysis transparency.

Data will be assessed independently by at least two people (or person/machine combination) with a process to resolve differences.

Additional information will **not** be sought from study investigators if required information is unclear or unavailable in the study publications/reports.

Reporting bias assessment

Risk of bias due to missing results will not be assessed

Certainty assessment

Certainty of findings will not be assessed

OUTCOMES TO BE ANALYSED

Main outcomes

Machine learning model performance metrics for emotion recognition tasks: accuracy, precision, recall, F1-score, and area under the curve (AUC). Neural network architectures employed (e.g., CNN types, traditional ML models). Datasets utilized for training and validation with their characteristics and representativeness limitations. Input features and variables used for emotion recognition (facial landmarks, action units, microexpressions, multimodal signals). Technical and methodological limitations impeding mobile implementation in resource-limited settings. All outcomes measured as reported by study authors across heterogeneous experimental conditions. No meta-analysis planned; outcomes synthesized through qualitative narrative synthesis with descriptive statistics.

Additional outcomes

Preprocessing and optimization techniques employed: data augmentation methods, class balancing techniques (SMOTE, oversampling, undersampling), hyperparameter optimization strategies (k-fold cross-validation, grid search, random search, Bayesian optimization). Model complexity characteristics: ensemble versus non-ensemble architectures, unimodal versus multimodal approaches. Implementation requirements: connectivity dependencies, offline functionality, computational requirements for mobile deployment. Dataset accessibility: public versus proprietary datasets. Population characteristics: age ranges within pediatric populations, diagnostic criteria used, sample sizes, geographic/cultural contexts.

PLANNED DATA SYNTHESIS

Strategy for data synthesis

No formal data synthesis is planned - data will be described but not combined.

CURRENT REVIEW STAGE

Stage of the review at this submission 1 change

| Review stage                                        | Started | Completed |
|-----------------------------------------------------|---------|-----------|
| Pilot work                                          | ✓       | ✓         |
| Formal searching/study identification               | ✓       | ✓         |
| Screening search results against inclusion criteria | ✓       | ✓         |
| Data extraction or receipt of IPD                   | ✓       | ✓         |
| Risk of bias/quality assessment                     | ✓       | ✓         |
| Data synthesis                                      | ✓       | ✓         |

Review status

The review is completed.

Publication of review results

Results of the review will be published in English.

### Review team members

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No conflict of interest declared.

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No conflict of interest declared.

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No conflict of interest declared.

### Named contact

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### Review affiliation

Universidad San Ignacio de Loyola

### Funding source

Review has no specific/external funding but is supported by guarantor/review team (non-commercial) institutions.

#### *Additional information about funding*

Supported by Universidad San Ignacio de Loyola as part of thesis research.

### Peer review

Protocol reviewed by thesis advisor (Professor with expertise in assistive technologies and ASD interventions) as part of academic thesis supervision process at Universidad San Ignacio de Loyola.

## ADDITIONAL INFORMATION

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### Additional information

This systematic review forms part of a thesis project focused on developing culturally adapted, accessible emotion recognition technologies for children with Autism Spectrum Disorder in Latin American contexts. While the review synthesizes global evidence on machine learning and mobile applications for emotion recognition, particular emphasis is placed on identifying implementation barriers relevant to resource-limited settings. The review follows PRISMA 2020 guidelines and employed dual independent review at all stages (screening, data extraction, quality assessment) with inter-rater reliability monitoring (Cohen's kappa >0.80). Complete supplementary materials including search strategies, PRISMA flow diagram, and quality assessment data will be made available upon publication.

### Review conflict of interest

Declared individual interests are recorded under team member details.. No additional interests are recorded for this review.

### Medical Subject Headings

Algorithms; Area Under Curve; Autism Spectrum Disorder; Child; Deep Learning; Emotions; Facial Recognition; Humans; Latin America; Mobile Applications; Resource-Limited Settings; Smartphone

### Revision note 1 change



Correcting current review stage fields that were not properly saved during initial registration. All review stages (data extraction, risk of bias assessment, and data synthesis) have been completed as of October 2025. Updated status from "planned or ongoing" to "completed" to accurately reflect the actual state of the systematic review.

## SIMILAR REVIEWS

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### Check for similar records already in PROSPERO

PROSPERO identified a number of existing PROSPERO records that were similar to this one (last check made on 7 December 2025). These are shown below along with the reasons given by that the review team for the reviews being different and/or proceeding.

- Emotion Recognition in Spoken Conversations: A Systematic Review and Meta-Analysis of Artificial Intelligence Approaches [published 23 April 2023] [CRD42023416879]. **The authors did not check this review**
- A Systematic Literature Review for Multi Class Arrhythmia Detection using Traditional Machine Learning Models [published 2 December 2025] [CRD420251245908]. **The authors did not check this review**
- The performance of machine learning models in predicting children with Autism Spectrum Disorder: A meta-analysis and systematic review [published 9 December 2024] [CRD42024619503]. The review was acknowledged as **similar** but the authors opted to continue because *there are differences in intervention or comparator, the review looks at additional or different outcomes, the review will be more up to date*. The authors also noted: "Our review focuses on emotion recognition as a therapeutic intervention tool rather than ASD diagnostic prediction, and uniquely addresses mobile deployment and resource-limited settings."
- Machine learning applications in diagnosing pediatric autism spectrum disorder: a systematic review and meta-analysis [published 16 July 2025] [CRD420251107034]. The review was acknowledged as **similar** but the authors opted to continue because *there are differences in intervention or comparator, the review looks at additional or different outcomes, the review will be more up to date*. The authors also noted: "Our review uniquely focuses on implementation barriers in resource-limited settings, offline functionality requirements, and accessibility considerations for Latin American contexts, which are not addressed in the comparator review."
- Intervention for teaching emotion recognition to children and adolescents with autism spectrum disorder: A Systematic Review [published 23 February 2025] [CRD420250654775]. The review was acknowledged as **similar** but the authors opted to continue because *there are differences in intervention or comparator, the review looks at additional or different outcomes*. The authors also noted: "Our review addresses technology specifications, ML model performance, and mobile implementation challenges rather than pedagogical intervention techniques."

### PROSPERO version history 1 change

- [Version 1.1, published 11 Dec 2025](#)
- [Version 1.0, published 11 Dec 2025](#)

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